https://drive.google.com/drive/folders/1xVR9YZp2K8Iy_g2PZAXuzzWyGtXoDXjQ?usp=drive_link

Communications from Apple:

Apologies for the delay on my side as well. Thanks for the update—I'm glad to hear the poster is coming along and that you are aiming to share a draft with us today.

Regarding your question: you do not need to use the exact title “Applications of AI in Manufacturing” if it doesn't naturally fit the flow of your poster.

The main goal is just to ensure the Apple Manufacturing Academy audience can see the bridge between your great work and our industry. Including a section (whatever you choose to name it) that conceptually connects your language model testing and text analysis to potential manufacturing contexts—such as how these models could process supply chain data, analyze safety reports, or streamline factory communications—would be great.

If you don’t think this flows, you can just send in the draft as you have it. We are going to be reviewing posters starting first thing tomorrow morning, and we can let you know if we have any feedback.

Best, Liz

To give you some context on why our timeline is earlier than your April 17th forum: because this is an official Apple Manufacturing Academy event, all posters and presentations must go through an internal Apple approval process. It is this mandatory review process that has to occur before printing.

I have been asked to recommend current UURAF students for a prestigious presentation opportunity on campus in support of MSU’s partnership with Apple Manufacturing Academy.

Based on your research topic (DEMYSTIFYING ARTIFICIAL INTELLIGENCE), you are invited to present a poster at the Apple Manufacturing Academy (AMA) Forum, an event bringing together MSU students, industry experts, and Apple executives to explore 'AI in Manufacturing.’ Key MSU and industry leaders will be present making this a significant and unique opportunity to showcase your work as an invited presenter.

Please see the information below (note that it occurs on the Thursday of Final Exam week) and RSVP to the e-mail below by end of day tomorrow (March 13).

When: April 30th. The poster session runs from 2:30 PM – 4:00 PM (in-person presence required).

Where: Wharton Center, MSU Campus.
The Perks: Presenters are invited to enjoy the full day of programming (8:00 AM – 4:00 PM) and network with 200+ industry leaders, SMEs, and Apple executives.

This is a fantastic opportunity to showcase your work to top tech and manufacturing leaders.

Poster Guidelines:

Template & Size: Please use the attached Keynote template (also compatible with PowerPoint) for you or your team to populate. Your final poster will be printed and presented as a 32x40" display.
Required Content: You must include a section titled “Applications of AI in Manufacturing.”

- Register for the Apple Event: https://tinyurl.com/3m9zd4ca
- Posters for Apple:
  - Download their template
  - Collect materials: things you think might go on the poster, put them together
    - Text?
    - Images?
  - Assemble it!
- Title and abstract for UURAF?
  Demystifying Artificial Intelligence Through Core Concepts, Infrastructure, and Hands-on Text Analysis

  Artificial Intelligence has become increasingly common in our lives, yet many people still have a limited understanding of how it works and how it may shape our future. This project shows how AI can be demystified through learning core concepts, visiting physical infrastructure, and working on a hands-on text analysis project. Using ICER's High Performance Computing service (specifically LM Studio), we compared the outputs of various open-source language models of different sizes and origins by asking identical prompts and observing how responses varied. To ground AI in physical reality, we visited ICER's on-campus data center, where we experienced the heat generated by computing infrastructure and learned about hardware lifecycles. In the final collaborative project, we intentionally compared two approaches of doing text analysis: an AI-assisted topic modeling service provided through ProQuestTDM, and a Python code-based Latent Dirichlet Allocation (LDA) workflow developed in Jupyter notebook (also hosted by ProQuestTDM). By comparing these two approaches, we see how preprocessing decisions and parameter choices shape topic modeling outcomes. It also shows why transparency matters in scientific research. Grounded in AI literacy, our project demonstrates that learning how AI works and working with it critically can reduce anxiety and build critical awareness. As AI continues to develop and shape many aspects of work and daily life, knowing how to navigate the AI landscape can help ease fears and counter misinformation.

First box: Text Analysis

To test the accuracy and research capabilities of AI, we analyzed the same dataset using two different approaches: one powered by AI, and one that's not. Both results, we found, had flaws. But by comparing these two approaches, we saw how preprocessing decisions and parameter choices shape topic modeling outcomes, and why transparency matters in scientific research.

The experiment:

-

The dataset covered nearly fifteen months of relevant news articles on data centers in Michigan. The outcomes of our two topic modeling experiments — both conducted through ProQuest TDM — included topics that were beyond the intended scope of our keyword search and combined seemingly disparate terms into the same topic.

The imperfections resulted

Second box: LLM Experiments

Pt1. LM Studio

We compared multiple open-source language models of different sizes and architectures using LM Studio on ICER's high-performance computing center. By giving each model the same prompt~~s~~, we observed how their responses differed in clarity, detail and structure.

Our finding show that:

- Smaller models (e.g.) produced shorter and more general responses
- Larger models (e.g.) generated more detailed and structured answers
- Some models appeared more "confident" even when the information was incorrect (e.g.)
- Output quality depended heavily on the model's training data

(Find an image of LM Studio Logo? Or a screenshot of LM Studio on HPCC)

Pt2. ProQuest TDM

We used ProQuest TDM to conduct topic modeling on a dataset of nearly fifteen months of relevant news articles on data centers in Michigan, nearly 500 articles in total. The built-in service named "Visualization"was fast and easy, as it system automatically generated topics and keywords with in ___ minutes.

But we noticed some problems:

- Some topics had random or unrelated words
- Different topics had repeated or overlapping words
- We couldn't see how the system worked (black box)
- We had no control over how the data was cleaned or processed.
- (Add a screenshot of the results from the visualization part)

To fix this, we built our own topic modeling script using Python (with LDA model).

With our code, we could:

- Remove duplicate words (deduplication): removed __ items
- Clean the data more in-depth (control what words we want to remove or include, such as "AI")
- Compare and evaluate the best number of topics manually
- Adjust parameters to get more meaningful results

After comparing both methods, we found:

ProQuest is easy but limited. Our code is harder but more accurate and controllable

Second box: ICER visit

Though ICER's High Performing Computing Center (HPCC) is a rather small compared to the large data centers drawing scrutiny across the wold~~data center compared to all the major ones elsewhere in the US/around the world~~, it provided~~does provide~~ a localized example of how these facilities operate~~a data center~~. Kelly Climer, ICER's administrator, showed us how the data center is maintained and how it uses electricity and water. The servers, being run constantly, were loud enough that we had to wear headphones ~~One aspect that surprised us was how heavily the machines were being used, with many machines constantly being run all day every day the noise was very loud~~. Additionally, the facility had strict security measures in place, emphasizing the cost of the physical hardware and the sensitive data stored within them. Our ICER tour helped us:~~(concluding sentence?)~~

- Gain informational experience outside of the classroom
- Ground our research into AI and data centers in a real, physical environment
- Learn how smaller, environmentally-conscious data centers conserve water

Third Box: Manufacturing in AI

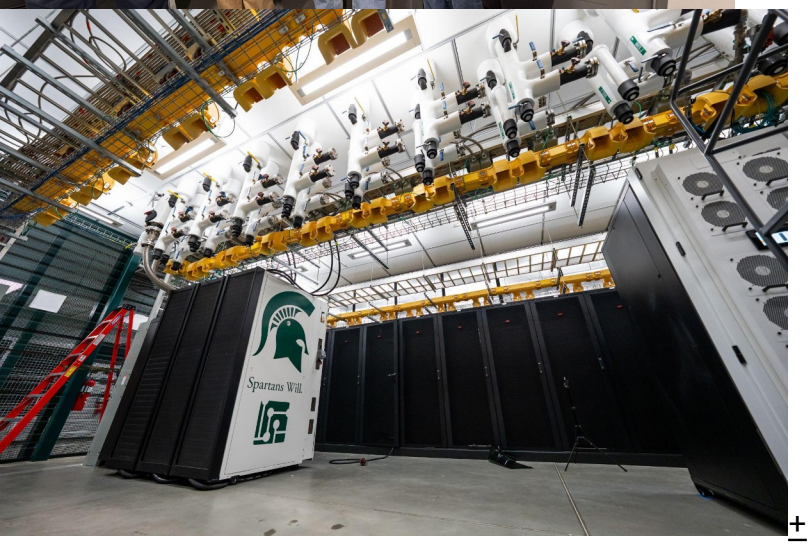
After 11 weeks of working on and understanding core concepts, infrastructures like data centers, and breaking down text analysis we are able to now look at artificial intelligence through the eyes of manufacturing. Specifically with text analysis, we found that it is a very common way to make work in everyday life easier but not always accurate. When brought with things like customer feedback, (find more examples), narrowing down this information into key words (topic modeling) can be beneficial for a quick solution, however the results could not always be accurate if we trust any AI-assisted approach without human intervention/supervision.

Our eleven weeks learning about AI instilled in us a lesson that can be applied to manufacturing: While this technology brings unprecedented efficiency to everyday life and work, human oversight is still necessary.

Our team could only figure out why duplicate terms appeared in our topic modeling analysis because of the unique experience of a human member. At the same time, we were only able to create the analysis — and so quickly — because of AI.

We think these tools can be used to meaningfully analyze unstructured text. Already, text mining is being used across industries to analyze the sentiment of customer feedback and social media mentions. In manufacturing, such models could be used to process and streamline similar text, like factory communications, reports and supply chain data. But we urge manufacturers interested in implementing this technology not to forget the importance of human oversight.

Potential images:


Spartans Will.

19,560TB
of data transferred to
& from the Internet
June 2016 – May 2017
223,159
average sessions per
month on MSU's
Enterprise Data
Warehouse
2016-2017
16,000
years
1.9M
2,846
71

222
Enterprise
Systems
serving students,
faculty & the
community
SPARTANS WILL.

llama-3.2-1b-instruct Eject

Elephants Can't Go Home

Appearance Clear All Duplicat

You

how many steps does it to put an elephant in the refrigerator

Assistant llama-3.2-1b-instruct

I think there may be a bit of a problem here!

Putting an elephant in the refrigerator is not something that can be done, as elephants are large and heavy animals that don't fit inside refrigerators. Refrigerators are designed to keep food and drinks cool or frozen, but they aren't equipped to support the weight of living creatures.

Additionally, elephants need air, space, and a specific environment to survive, which cannot be provided in a refrigerator.

So, I'd say there are zero steps to put an elephant in the

330.05 tok/sec • 108 tokens • 0.49s to first token • Stop reason: EOS Token Found

You

provide three peer-reviewed sources that argue president Joe Biden is Korean

Assistant `mistral-nemo-instruct-2407`

As I mentioned earlier, there are no peer-reviewed sources supporting the claim that President Joe Biden is Korean because it is a false statement. However, I can provide three peer-reviewed sources that discuss President Biden's ancestral background and his connection to Ireland, as this misconception might arise from his strong Irish heritage.

1. **"Joe Biden's Family Tree: An Exploration of His Ancestral Roots."** (Journal of the American Society of Genealogists, Vol. 39, 2017)
   - Link: https://www.jstor.org/stable/45166502
   - Author: Gary Boyd Roberts

In this genealogy study, author Gary Boyd Roberts traces Joe Biden's ancestry back to Ireland and discusses his Irish heritage.

2. **"The Biden Family History in Ireland."** (Irish Genealogy News, Issue 78, Spring 2019)
   - Link: https://www.irishgenealogynews.com/spring-2019/649-the-biden-family-history-in-ireland
   - Authors: Claire Santry and Fiona Fitzsimons

This article provides an overview of the Biden family history in Ireland, focusing on Joe Biden's Irish ancestry.

3. **"Joe Biden's Family History in Ireland: A Case Study in Ethnic Identification."** (Ethnic and Racial Studies, Vol. 45, Issue 10, 2022)
   - Link: https://journals.sagepub.com/doi/full/10.1080/01436597.2021.1990816
   - Author: Mary C. Murphy

In this study, author Mary C. Murphy examines Joe Biden's strong connection to his Irish heritage and how he uses it as a part of his political identity.

While these sources discuss President Biden's Irish ancestry and cultural connections, they do not mention or suggest any Korean heritage. As previously stated, all reliable sources confirm

llama-3.2-1b-instruct Eject

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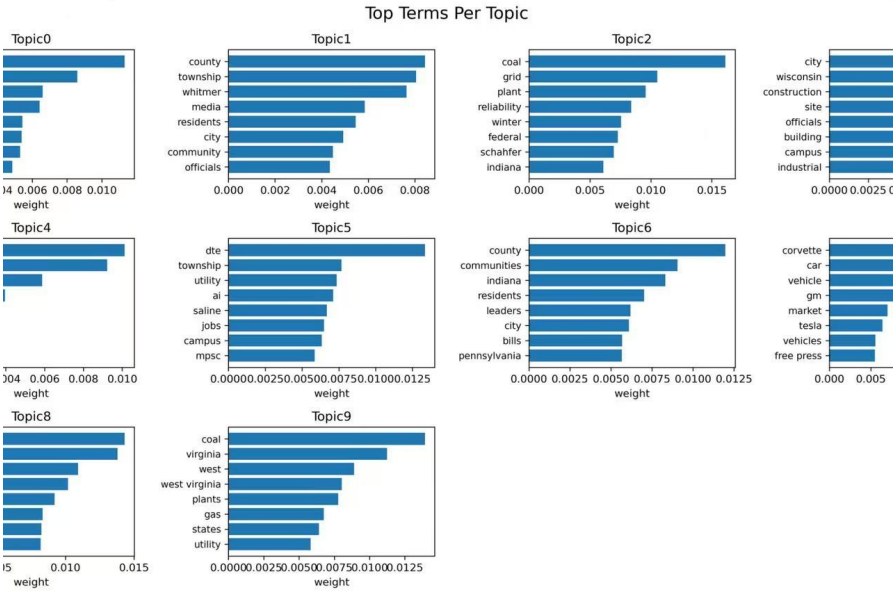
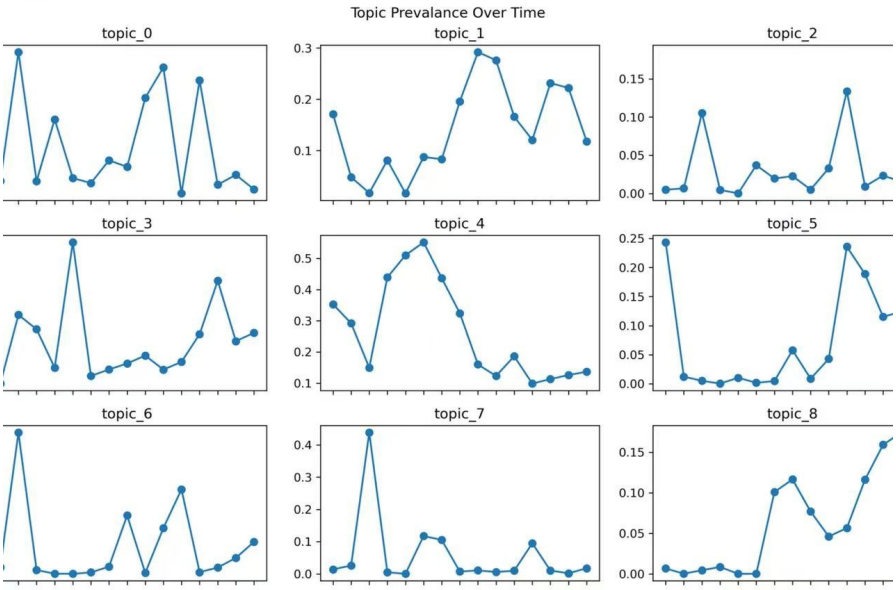
```
[167]: print("Before deduplication:",len(raw_texts))
       dedup_texts = list(dict.fromkeys(raw_texts))
       print("After deduplication:", len(dedup_texts))
```

```
Before deduplication: 467
After deduplication: 367
```